

Assignment No: 4

Class and Object, Constructor, this and static

1. Write a program to print the average of three numbers entered by user by creating a class named 'Average' having a method to calculate and print the average. Define another driver class to demonstrate the basic operation.

SOLUTION:

```
import java.util.Scanner;
```

```
class Average {
```

```
    private double num1, num2, num3;
```

```
//CONSTRUCTOR
```

```
    public Average(double num1, double num2, double num3) {
```

```
        this.num1 = num1;
```

```
        this.num2 = num2;
```

```
        this.num3 = num3;
```

```
    }
```

```
    public void calculateAndPrintAverage() {
```

```
        double average = (num1 + num2 + num3) / 3;
```

```
        System.out.println("Average of the three numbers: " + average);
```

```
    }
```

```
}
```

```
// DRIVER CLASS
```

```
public class AverageDemo {
```

```
    public static void main(String[] args) {
```

```
        Scanner scanner = new Scanner(System.in);
```

```
        System.out.println("Enter three number:");
```

```
        double a = scanner.nextDouble();
```

```
        double b = scanner.nextDouble();
```

```
        double c = scanner.nextDouble();
```

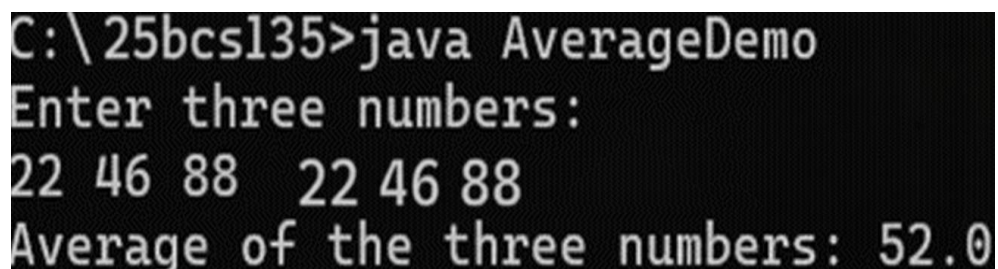
```
        Average averageCalculator = new Average(a, b, c);
```

```
        averageCalculator.calculateAndPrintAverage();
```

```
        scanner.close();
```

```
    }
```

```
}
```



```
C:\25bcs135>java AverageDemo
Enter three numbers:
22 46 88 22 46 88
Average of the three numbers: 52.0
```

2. Write a program to create a class named 'Student' with members 'name', 'roll_no', 'branch'. Declare two methods to input the student details and display the details of the student. Create a driver class to test the functionalities of the above class.

SOLUTION:

```
import java.util.Scanner;

class Student {
    String name;
    int roll_no;
    String branch;

    void inputDetails() {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter student name: ");
        name = sc.nextLine();
        System.out.print("Enter roll number: ");
        roll_no = sc.nextInt();
        sc.nextLine();
        System.out.print("Enter branch: ");
        branch = sc.nextLine();
    }

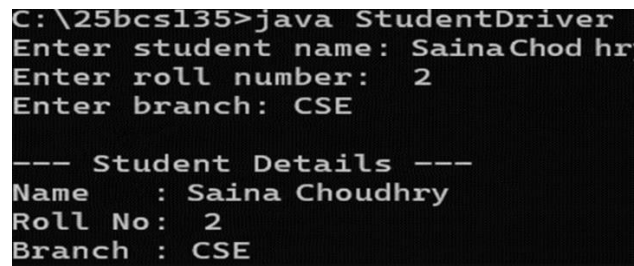
    void displayDetails() {
        System.out.println("\n--- Student Details ---");
        System.out.println("Name : " + name);
        System.out.println("Roll No: " + roll_no);
        System.out.println("Branch : " + branch);
    }
}

public class StudentDriver {
    public static void main(String[] args) {
        Student s1 = new Student();
        s1.inputDetails();
    }
}
```

```

        s1.displayDetails();
    }
}

```



```

C:\25bcs135>java StudentDriver
Enter student name: SainaChod hr
Enter roll number: 2
Enter branch: CSE

--- Student Details ---
Name    : Saina Choudhry
Roll No: 2
Branch  : CSE

```

3. Declare a class Complex with member real and imaginary part. Define a method initialise() to input the two complex numbers, show() to display the complex number and add() to add the two complex numbers. Declare another class to illustrate the operations of the Complex class.

SOLUTION:

```

import java.util.Scanner;

class Complex {
    int real;
    int imaginary;

    void initialise() {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter real part: ");
        real = sc.nextInt();
        System.out.print("Enter imaginary part: ");
        imaginary = sc.nextInt();
    }

    void show() {
        if (imaginary >= 0) {
            System.out.println(real + " + " + imaginary + "i");
        } else {
            System.out.println(real + " - " + (-imaginary) + "i");
        }
    }
}

Complex add(Complex c1, Complex c2) {

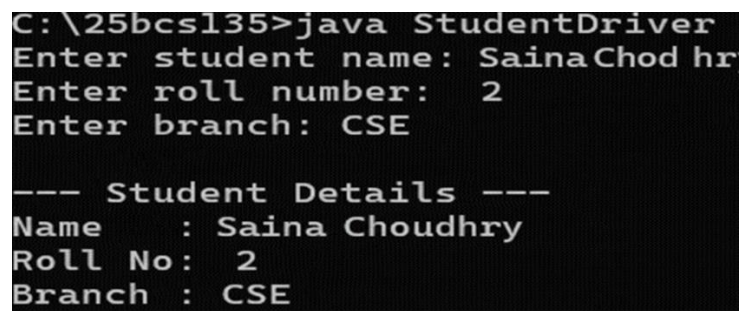
```

```

        Complex temp = new Complex();
        temp.real = c1.real + c2.real;
        temp.imaginary = c1.imaginary + c2.imaginary;
        return temp;
    }
}

public class ComplexDriver {
    public static void main(String[] args) {
        Complex c1 = new Complex();
        Complex c2 = new Complex();
        System.out.println("Enter first complex number:");
        c1.initialise();
        System.out.println("Enter second complex number:");
        c2.initialise();
        System.out.print("First Complex Number: ");
        c1.show();
        System.out.print("Second Complex Number: ");
        c2.show();
        Complex sum = new Complex();
        sum = sum.add(c1, c2);
        System.out.print("Sum of Complex Numbers: ");
        sum.show();
    }
}

```



A screenshot of a terminal window showing the execution of a Java program. The prompt is 'C:\25bcs135>java StudentDriver'. The program prompts for 'Enter student name:', 'Enter roll number:', and 'Enter branch:'. The user enters 'SainaChod hr', '2', and 'CSE' respectively. The program then displays '--- Student Details ---' followed by 'Name : Saina Choudhry', 'Roll No: 2', and 'Branch : CSE'.

4. Write a program to print the area of a rectangle by creating a class named 'Area' having two methods. First method named as 'setDim ()' takes length and breadth of

rectangle as parameters and the second method named as 'getArea ()' returns the area of the rectangle. Length and breadth of rectangle are entered through keyboard. [use 'this' keyword]

SOLUTION:

```
import java.util.Scanner;

class Area {

    private int length;

    private int breadth;

    // Method

    public void setDim(int length, int breadth) {

        this.length = length;

        this.breadth = breadth;

    }

    // Method to calculate and return area

    public int getArea() {

        return this.length * this.breadth;

    }

}

public class AreaRectangle {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter length of rectangle: ");

        int length = sc.nextInt();
```

```

        System.out.print("Enter breadth of rectangle: ");

        int breadth = sc.nextInt();

        //object

        Area rect = new Area();

        rect.setDim(length, breadth);

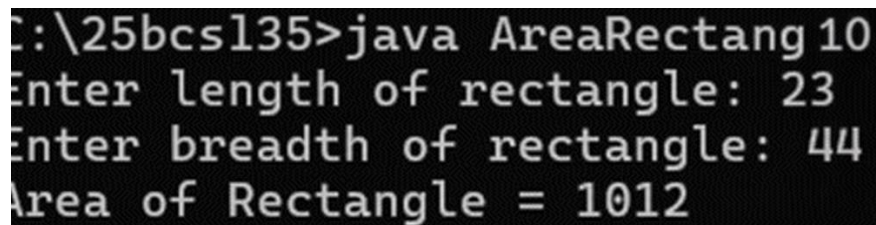
        System.out.println("Area of Rectangle = " + rect.getArea());

        sc.close();

    }

}

```



```

C:\25bcs135>java AreaRectang 10
Enter length of rectangle: 23
Enter breadth of rectangle: 44
Area of Rectangle = 1012

```

- Write a program to declare a class employee with data members empid, ename, dept, sal. Add a method getData(), putData() to accept and display the details of n employees. Define a driver class to create the objects of employee class and perform basic operations.

SOLUTION:

```

import java.util.Scanner;

class Employee{

    int empid;

    String ename;

    String dept;

```

```

        double sal;

        public void getData(Scanner sc){

            System.out.print("Enter Employee ID: ");

            empid = sc.nextInt();

            sc.nextLine();

            System.out.print("Enter Employee Name: ");

            ename = sc.nextLine();

            System.out.print("Enter Department: ");

            dept = sc.nextLine();

            System.out.print("Enter Salary: ");

            sal = sc.nextDouble();

        }

// Method

    public void putData() {

        System.out.println("Employee ID : " + empid);

        System.out.println("Employee Name : " + ename);

        System.out.println("Department : " + dept);

        System.out.println("Salary : " + sal);

        System.out.println("-----");

    }

}

public class EmployeeDetails{

```



```
public static void main(String[] args) {  
  
    Scanner sc = new Scanner(System.in);  
  
    System.out.print("Enter number of employees: ");  
  
    int n = sc.nextInt();  
  
    Employee[] emp = new Employee[n];  
  
    for (int i = 0; i < n; i++) {  
  
        System.out.println("\nEnter details for Employee " + (i + 1) + ":");  
  
        emp[i] = new Employee();  
  
        emp[i].getData(sc);  
  
    }  
  
    System.out.println("\nEmployee Details");  
  
    for (int i = 0; i < n; i++) {  
  
        System.out.println("Employee " + (i + 1) + ":");  
  
        emp[i].putData();  
  
    }  
  
  
    sc.close();  
  
    }  
}
```

```

Center number of employees: 3
Enter details for Employee 1:
Enter Employee ID: 1001
Enter Employee Name: XYZ
Enter Department: Architecture
Enter Salary: 100000

Enter details for Employee 2:
Enter Employee ID: 3900
Enter Employee Name: Annabelle
Enter Department: Artist
Enter Salary: 10000000

Enter details for Employee 3:
Enter Employee ID: 2100
Enter Employee Name: Mickey
Enter Department: Sales
Enter Salary: 2100000

Employee Details
Employee 1:
Employee ID      : 1001
Employee Name    : XYZ
Department       : Architecture
Salary           : 100000.0
-----
Employee 2:
Employee ID      : 3900
Employee Name    : Annabelle
Department       : Artist
Salary           : 1.00E
-----
Employee 3:
Employee ID      : 2100
Employee Name    : Mickey
Department       : Sales
Salary           : 2100000.0

```

6. Define a class Student having the attribute regNo, stdName, branch and CGPA. Write 2 methods to accept and display the student details. Read the details of 50 students using an array of Student class object. Display the student details who has secured the highest CGPA.

SOLUTION:

```
import java.util.Scanner;
```

```
class Student {
```

```
    int regNo;
```

```
    String stdName;
```

```
    String branch;
```

```
    double CGPA;
```

```
// Method

public void getData(Scanner sc) {

    System.out.print("Enter Registration Number: ");

    regNo = sc.nextInt();

    sc.nextLine();

    System.out.print("Enter Student Name: ");

    stdName = sc.nextLine();

    System.out.print("Enter Branch: ");

    branch = sc.nextLine();

    System.out.print("Enter CGPA: ");

    CGPA = sc.nextDouble();

}

public void putData() {

    System.out.println("Registration No : " + regNo);

    System.out.println("Student Name : " + stdName);

    System.out.println("Branch      : " + branch);

    System.out.println("CGPA      : " + CGPA);

}

}

public class StudentDetails{

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
```

```
System.out.print("Enter number of students (max 50): ");

int n = sc.nextInt();

if (n > 50) {

    System.out.println("Maximum limit is 50 students.");

    return;

}

Student[] students = new Student[n];

for (int i = 0; i < n; i++) {

    System.out.println("\nEnter details for Student " + (i + 1) + ":");

    students[i] = new Student();

    students[i].getData(sc);

}

Student topStudent = students[0];

for (int i = 1; i < n; i++) {

    if (students[i].CGPA > topStudent.CGPA) {

        topStudent = students[i];

    }

}

System.out.println("\nStudent with Highest CGPA");

topStudent.putData();

sc.close();
```

}

```
C:\25bcs115>java StudentDetails
Enter number of students (max 50): 2

Enter details for Student 1:
Enter Registration Number: 100200
Enter Student Name: Dekisugi
Enter Branch: CSE
Enter CGPA: 10

Enter details for Student 2:
Enter Registration Number: 100210
Enter Student Name: Sunio
Enter Branch: CSE
Enter CGPA: 9.5

Student with Highest CGPA
Registration No : 100200
Student Name    : Dekisugi
Branch          : CSE
CGPA            : 10.0
```

7. Define a class 'Box' that uses a parameterized constructor to initialize the dimensions of a box. The dimensions of the Box are width, height, depth. The class should have a method that can return the volume of the box. Create an object of the Box class and test the functionalities[use this keyword].

SOLUTION:

```
import java.util.Scanner;
```

```
class Box {
```

```
    private double width;
```

```
    private double height;
```

```
    private double depth;
```

```
    // Parameterized constructor
```

```
    public Box(double width, double height, double depth) {
```

```
        this.width = width;
```

```
        this.height = height;
```

```
        this.depth = depth;
    }
// Method

    public double getVolume() {
        return this.width * this.height * this.depth;
    }
}

public class BoxArea {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter width of the box: ");
        double w = sc.nextDouble();
        System.out.print("Enter height of the box: ");
        double h = sc.nextDouble();
        System.out.print("Enter depth of the box: ");
        double d = sc.nextDouble();
        //object of Box using parameterized constructor
        Box myBox = new Box(w, h, d);
        System.out.println("Volume of the Box = " + myBox.getVolume());
        sc.close();
    }
}
```

```
C:\255cs135>java BoxArea
Enter width of the box: 2
Enter height of the box: 4
Enter depth of the box: 5
Volume of the Box = 40.0
```

8. Design a class 'ComplexNum' to manipulate Complex numbers having data members as real and img. The class should have a parameterized constructor to initialize its data members. It should also have methods displayCompNumber() to display the complex number (in the format 5+3i for example), addCompNumber() to add two Complex numbers. Test these methods by creating main method in another class.

SOLUTION:

```
class ComplexNum {

    private int real;

    private int img;

    // Parameterized constructor

    public ComplexNum(int real, int img) {

        this.real = real;

        this.img = img;

    }

    // Method

    public void displayCompNumber() {

        if (img >= 0)

            System.out.println(real + " + " + img + "i");

        else
```

```

        System.out.println(real + " - " + (-img) + "i");
    }

    // Method to add two complex numbers

    public ComplexNum addCompNumber(ComplexNum c) {

        int newReal = this.real + c.real;

        int newImg = this.img + c.img;

        return new ComplexNum(newReal, newImg);

    }
}

public class Complexadd {

    public static void main(String[] args) {

        // two complex numbers using parameterized constructor

        ComplexNum c1 = new ComplexNum(5, 3);

        ComplexNum c2 = new ComplexNum(2, -4);

        System.out.print("First Complex Number: ");

        c1.displayCompNumber();

        System.out.print("Second Complex Number: ");

        c2.displayCompNumber();

        ComplexNum sum = c1.addCompNumber(c2);

        System.out.print("Sum of the Complex Numbers: ");

        sum.displayCompNumber();

    }
}

```



```
}
```

```
C:\25bcs135>java Complexbsl35
First Complex Number: 5 + 3i
Second Complex Number: 2 - 4i
Sum of the Complex Numbers: 7 - 1i
```

9. Design a class 'Point' with data members as xCo and yCo. The class should have a parameterized constructor to initialize its data members. Define a method distanceBetPoints() which returns the distance between two points.

SOLUTION:

```
class Point {

    private double xCo;

    private double yCo;

    // Parameterized constructor

    public Point(double xCo, double yCo) {

        this.xCo = xCo;

        this.yCo = yCo;

    }

    // Method to calculate distance between two points

    public double distanceBetPoints(Point p) {

        double dx = this.xCo - p.xCo;

        double dy = this.yCo - p.yCo;

        return Math.sqrt(dx * dx + dy * dy);

    }

}
```

```

public class PointMain {

    public static void main(String[] args) {

        Point p1 = new Point(3, 4);

        Point p2 = new Point(7, 1);

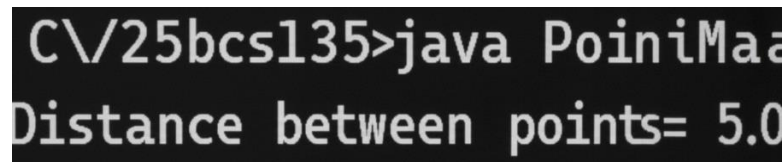
        double distance = p1.distanceBetPoints(p2);

        System.out.println("Distance between points = " + distance);

    }

}

```



```

C:\25bcs135>java PoiniMaa
Distance between points= 5.0

```

10. Design a class 'Time' having data members as hour, minute and second. The class should have a parameterized constructor to initialize its data members. It should also have methods displayTime() to display the time in HH:MM:SS format and addTime() to add two times. Test these methods by creating a main method in another class.

SOLUTION:

```

import java.util.Scanner;

class Time {

    private int hour;

    private int minute;

    private int second;

    // Parameterized constructor

    public Time(int hour, int minute, int second) {

        this.hour = hour;

```

```
        this.minute = minute;

        this.second = second;

        normalizeTime();

    }

    // Normalize time (adjust minutes and hours if needed)

    private void normalizeTime() {

        if (second >= 60) {

            minute += second / 60;

            second = second % 60;

        }

        if (minute >= 60) {

            hour += minute / 60;

            minute = minute % 60;

        }

        if (hour >= 24) {

            hour = hour % 24; // keep it in 24-hour format

        }

    }

    // Display time

    public void displayTime() {

        System.out.printf("%02d:%02d:%02d%n", hour, minute, second);

    }

}
```

```
public Time addTime(Time t) {  
  
    int newHour = this.hour + t.hour;  
  
    int newMinute = this.minute + t.minute;  
  
    int newSecond = this.second + t.second;  
  
    return new Time(newHour, newMinute, newSecond);  
  
}  
  
}  
  
public class TimeMain {  
  
    public static void main(String[] args) {  
  
        Scanner sc = new Scanner(System.in);  
  
        System.out.println("Enter first time:");  
  
        System.out.print("Hours: ");  
  
        int h1 = sc.nextInt();  
  
        System.out.print("Minutes: ");  
  
        int m1 = sc.nextInt();  
  
        System.out.print("Seconds: ");  
  
        int s1 = sc.nextInt();  
  
        System.out.println("\nEnter second time:");  
  
        System.out.print("Hours: ");  
  
        int h2 = sc.nextInt();  
  
        System.out.print("Minutes: ");  
  
        int m2 = sc.nextInt();  

```

```

        System.out.print("Seconds: ");

        int s2 = sc.nextInt();

        //Time objects

        Time t1 = new Time(h1, m1, s1);

        Time t2 = new Time(h2, m2, s2);

        System.out.print("\nFirst Time: ");

        t1.displayTime();

        System.out.print("Second Time: ");

        t2.displayTime();

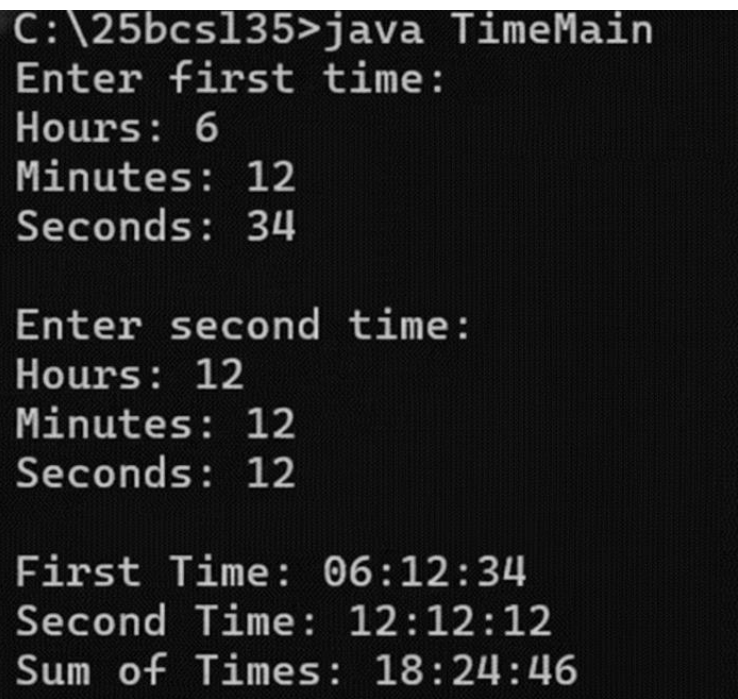
        Time sum = t1.addTime(t2);

        System.out.print("Sum of Times: ");

        sum.displayTime();

        sc.close();
    }
}

```



A screenshot of a terminal window showing the execution of a Java program. The prompt is 'C:\25bcs135>java TimeMain'. The program prompts for 'Enter first time:' and receives 'Hours: 6', 'Minutes: 12', and 'Seconds: 34'. It then prompts for 'Enter second time:' and receives 'Hours: 12', 'Minutes: 12', and 'Seconds: 12'. Finally, it displays the results: 'First Time: 06:12:34', 'Second Time: 12:12:12', and 'Sum of Times: 18:24:46'.

```

C:\25bcs135>java TimeMain
Enter first time:
Hours: 6
Minutes: 12
Seconds: 34

Enter second time:
Hours: 12
Minutes: 12
Seconds: 12

First Time: 06:12:34
Second Time: 12:12:12
Sum of Times: 18:24:46

```

